

CampusEats — Services, Contracts & Schema Design

1. Capabilities

CampusEats provides the following main capabilities:

- View and search food menu
- View food details
- Add food to cart
- Remove food from cart
- Place an order
- Make payment
- Cancel an order
- Track an order
- Add, update and remove food items
- Update food availability
- Manage users and vendors

2. Service Design

Catalogue Service

Responsible: [Ashish Kumar Singh]

Handles menus and food items.

Owns: Menus, Categories, Food Items

Order Service

Responsible: [Abhishek Kumar Jana]

Handles carts and food orders.

Owns: Carts, Cart Items, Orders, Order Items

Payment Service

Responsible: [Tushar Bharadwaj]

Handles payments for orders.

Owns: Payments

User & Admin Service

Responsible: [Bikash Pradhan]

Handles users and vendors.

Owns: Users, Vendors

Service Communication

Caller Service	Target Service	Operation
Order	Catalogue	checkItem
Order	Catalogue	getMenu
Order	Payment	processPayment
Order	User & Admin	getVendor

Services communicate through defined operations and do not directly access another service's data.

3. Service Contracts

Catalogue Service

Operation	Input	Output	Errors
getMenu	Menu request	Menu items	Menu not found
checkItem	Food item	Item details and availability	Item not found/unavailable
searchFood	Search text	Matching items	No results

Order Service

Operation	Input	Output	Errors
addToCart	Item, quantity	Updated cart	Item unavailable
removeFromCart	Item	Updated cart	Item not in cart
placeOrder	Cart, payment details	Order confirmation	Empty cart/payment failed
cancelOrder	Order	Cancellation result	Order not found

Payment Service

Operation	Input	Output	Errors
processPayment	Amount, payment details	Payment result	Payment failed
getPaymentStatus	Payment reference	Payment status	Payment not found

User & Admin Service

Operation	Input	Output	Errors
getVendor	Vendor request	Vendor details	Vendor not found
manageUser	User details	Updated user	Invalid user

4. Central Operation — placeOrder

placeOrder is used by a student to place an order from the items in the cart.

Input: Cart items and payment details.

Output: Order confirmation, total amount and payment status.

Main Steps

1. Check that the cart is not empty.
2. Check that the selected food items are available.
3. Get the vendor information.
4. Process the payment.
5. Create the order and return the confirmation.

Possible Errors

- Empty cart
- Food item unavailable
- Vendor not found
- Payment failed

The caller does not need to know how the order, payment or food data is stored internally.

5. Service Validation

Service	Reachable	Self-contained	Contract	Independent	Loosely Coupled
Catalogue	Yes	Yes	Yes	Yes	Yes
Order	Yes	Yes	Yes	Yes	Yes
Payment	Yes	Yes	Yes	Yes	Yes
User & Admin	Yes	Yes	Yes	Yes	Yes

The services have their own data and communicate through defined operations instead of directly accessing each other's data.